# Activity 3 / Document recommendations

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## Approach

We reused 32 simulated records from the earlier Aster prototype and wrote six policy/process records, for a self-contained corpus of 38 documents. The CSV stores ID, title, text and category. The Python program accepts a seed ID or free text and returns five ranked documents.

Text is lowercased and punctuation is replaced with spaces. TfidfVectorizer represents word unigrams and bigrams with English stopwords removed. Cosine similarity compares the seed or query vector with each document. Seed results exclude themselves; zero-overlap results are suppressed. This is a lexical baseline, not a semantic embedding model.

## Three executed cases

| Input | Top five, with cosine score |
|---|---|
| Policy / PP-01 | IR-01 .1999 · PP-03 .1427 · PP-06 .1207 · TG-01 .1165 · PP-02 .1046 |
| Incident / IR-01 | PP-01 .1999 · IR-04 .1665 · TP-01 .1279 · RN-01 .1206 · IR-06 .1030 |
| Free-text query | IR-01 .3323 · TP-01 .2674 · RN-01 .1604 · PP-01 .0921 · TG-01 .0781 |

Query: “How do we reproduce and fix the command display freeze after a network reconnect?” The full inputs, output and commentary are in evidencias/ as console logs and PNG captures.

# What the results tell us

A useful first pass, with visible limits

## Business utility

The policy seed leads to an incident, handoff and closure processes, and a recovery guide. The free-text question leads to the incident, reproduction procedure and release correction. Support can reuse past diagnoses; QA can start from a recorded test; Operations can check an approved guide. Shared references reduce duplicate investigation and make handoff decisions faster. A specialist must still verify the current version before acting.

## Areas for improvement

Template bias: IR-04 and IR-06 entered the incident Top-5 because tickets share standard wording, even though their symptoms differ. Reduce boilerplate weight and evaluate with human relevance labels.

Synonyms and ambiguity: terms such as sign-in and authentication may not overlap; a short query can refer to different systems. Compare this baseline with embeddings or a controlled vocabulary.

Production controls: add owner, system, version, review date and status; filter by user role before ranking; hide superseded guides; measure precision at five and collect resolution feedback. The small synthetic corpus is not evidence of live performance.

## Reflection video

The narrated reflection (2 minutes 29 seconds) answers how recommendations reduce rework, where they fail, and what a real deployment needs. The video file is included in this electronic folder as organizational_reflection.mp4.

Open the video in this electronic folder
Repository video link (active after publication)

Method and evidence: recommender.py · corpus.csv · evidencias/test1_policy, test2_incident, test3_query · video_script.txt